

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8      int dp[n];
9
10     for (int i = 0; i < n; i++) {
11         scanf("%d", &a[i]);
12         dp[i] = 1;
13     }
14
15     int maxLength = 1;
16
17     for (int i = 1; i < n; i++) {
18         for (int j = 0; j < i; j++) {
19             if (a[j] <= a[i]) {
20                 if (dp[j] + 1 > dp[i]) {
21                     dp[i] = dp[j] + 1;
22                 }
23             }
24         }
25         if (dp[i] > maxLength) {
26             maxLength = dp[i];
27         }
28     }
29
30     printf("%d\n", maxLength);
31
32     return 0;
33 }
```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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